



Designing A C.A.S.E. Air Quality Monitor

CPS: Industrial and Engineering
Technology

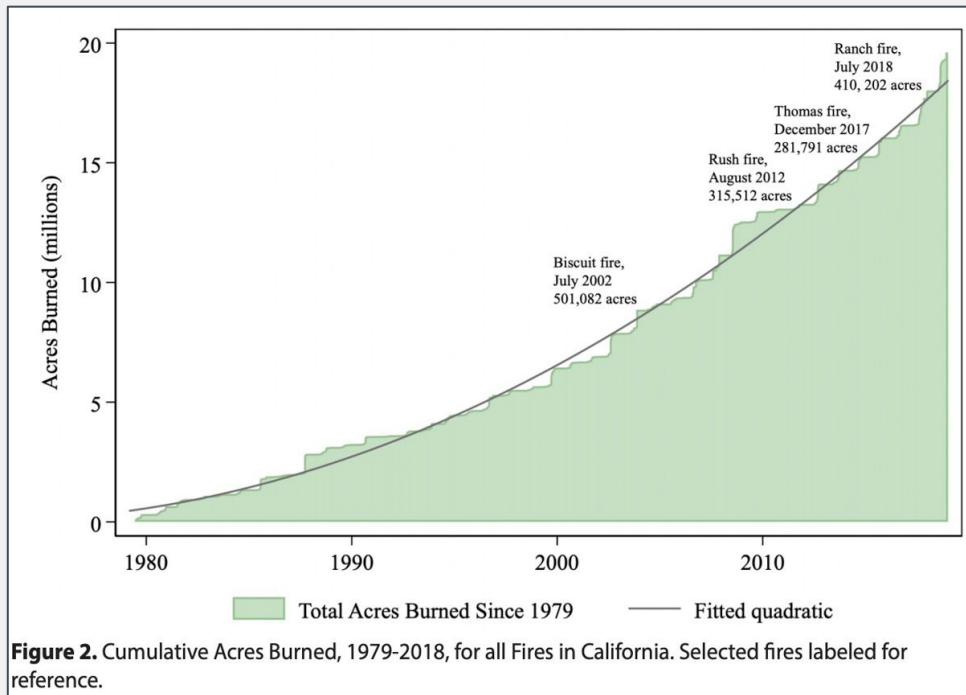


01

BACKGROUND

Problems, Health Implications,
Current Solutions, Aims

Background



Increasing trend of acre
damage overtime



Photo of wildfire in
La Crescenta, May 2022

Implications

- 34 million people in the US suffer from a chronic respiratory condition
 - **545 million people** worldwide
- Inhaling smoke and other harmful particulates can **exacerbate respiratory conditions**, even making them lethal
 - Residents in indigenous/underdeveloped areas are **6x more likely to die** from inhaling harmful particulates



Most Common Air Quality Monitors Used



AirVisual Outdoor

Weatherized outdoor air quality monitor with modular, field-swappable PM and CO₂ sensors.

Measures

- PM2.5
- PM1
- PM10
- CO₂
- Temperature
- Humidity

\$349



PurpleAir

A proven air quality monitoring solution for home enthusiasts and air quality professionals alike.

Measures

- PM2.5

\$229 + \$40 cable



Clarity Node-S

A sensing-as-a-service solution inclusive of solar and battery powered hardware, global cellular connectivity, and data assurance support for government agencies, schools, and community groups. Not intended for individual consumer purchases.

Measures

- PM2.5
- NO₂

\$501-1300

Problems

- Lack of awareness of the **harmful effects** of air pollution in developing countries
- Current Air Monitors
 - High Cost
 - No Device Warranty
 - Unknown Device Accuracy

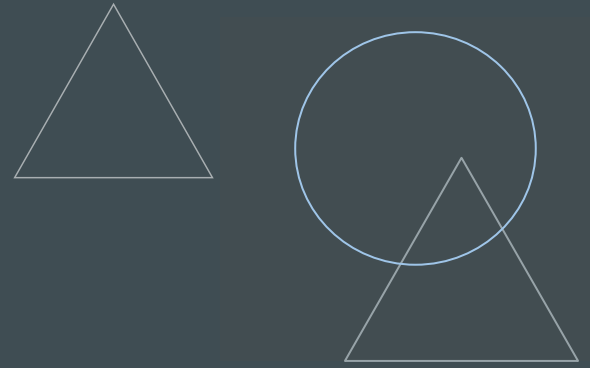
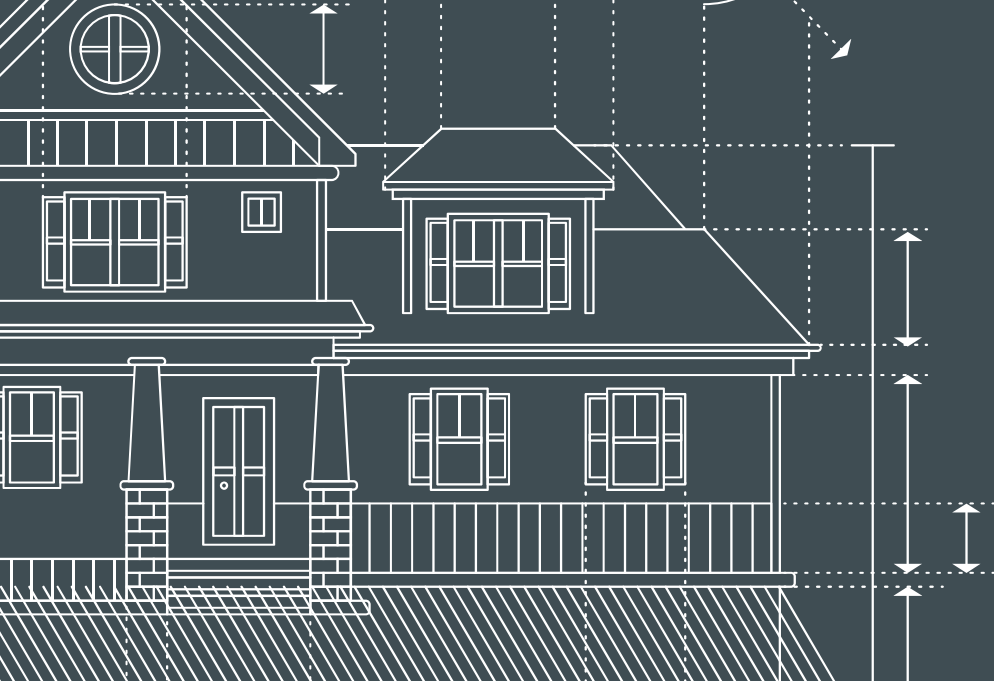
Disproportionately abundant monitors in industrialized areas



Project Aim

To create a consumer-friendly air quality monitor that is...

- **Cost** efficient
- **Accuracy** equivalent or better than current air quality monitors
- **Self-sustaining** power source (renewable)
- **Easy-to-use**



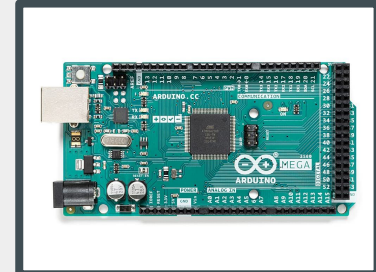
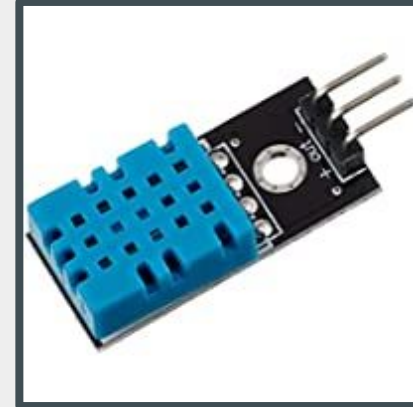
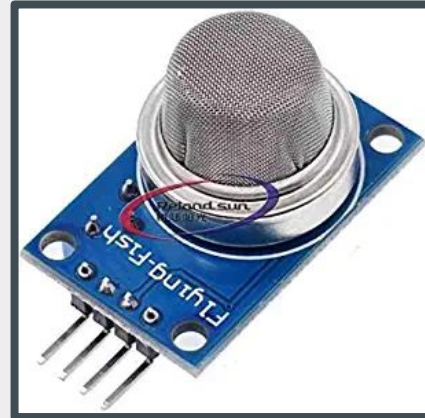
02

RESEARCH

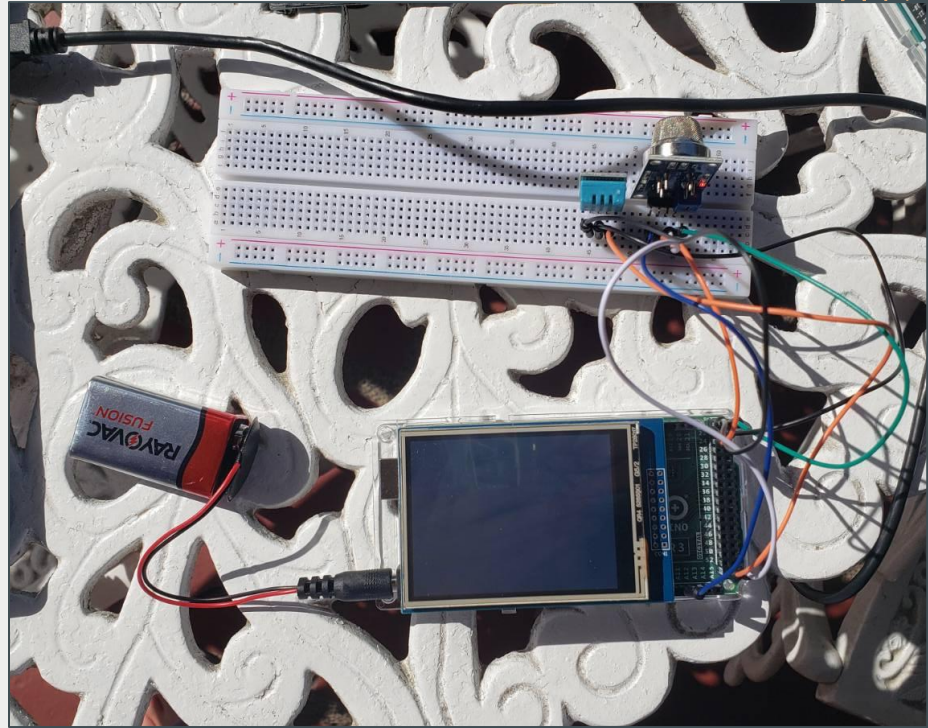
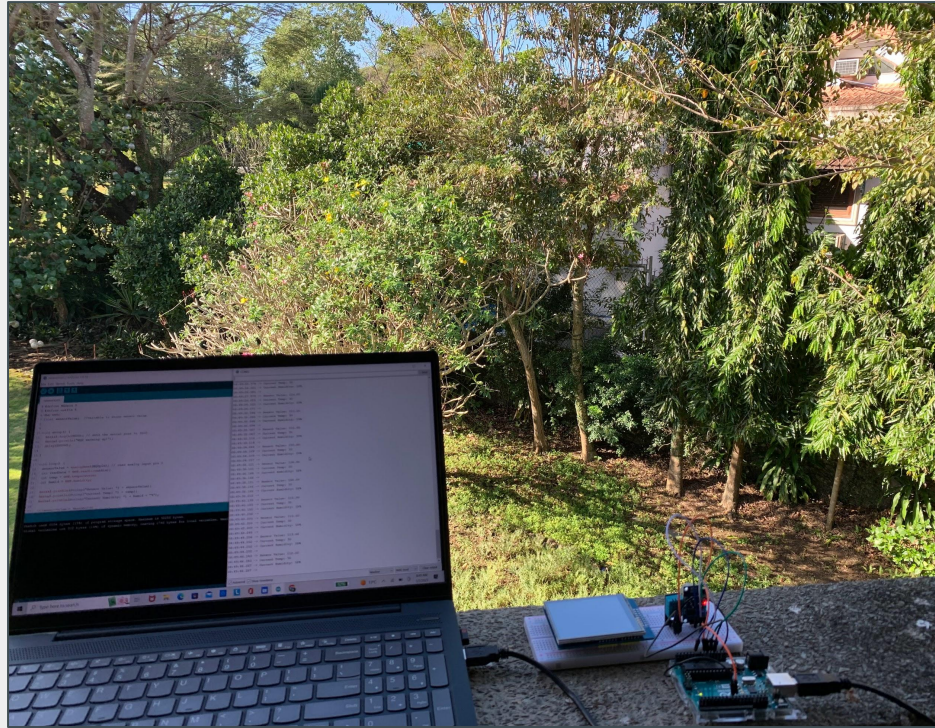
Set-up, Design, Test!

Materials

- PMS5003 particle concentration sensor – PM1.0, **2.5**, 10.0
- DHT11 sensor – humidity, temp
- MQ2 smoke sensor
- 6V solar cell
- 3.7V rechargeable lithium battery
- Arduino Mega



Hardware



Button Information + Capabilities•

- 3 main buttons (**bold + underlined**)
 - Info displayed from button in bullet points

General

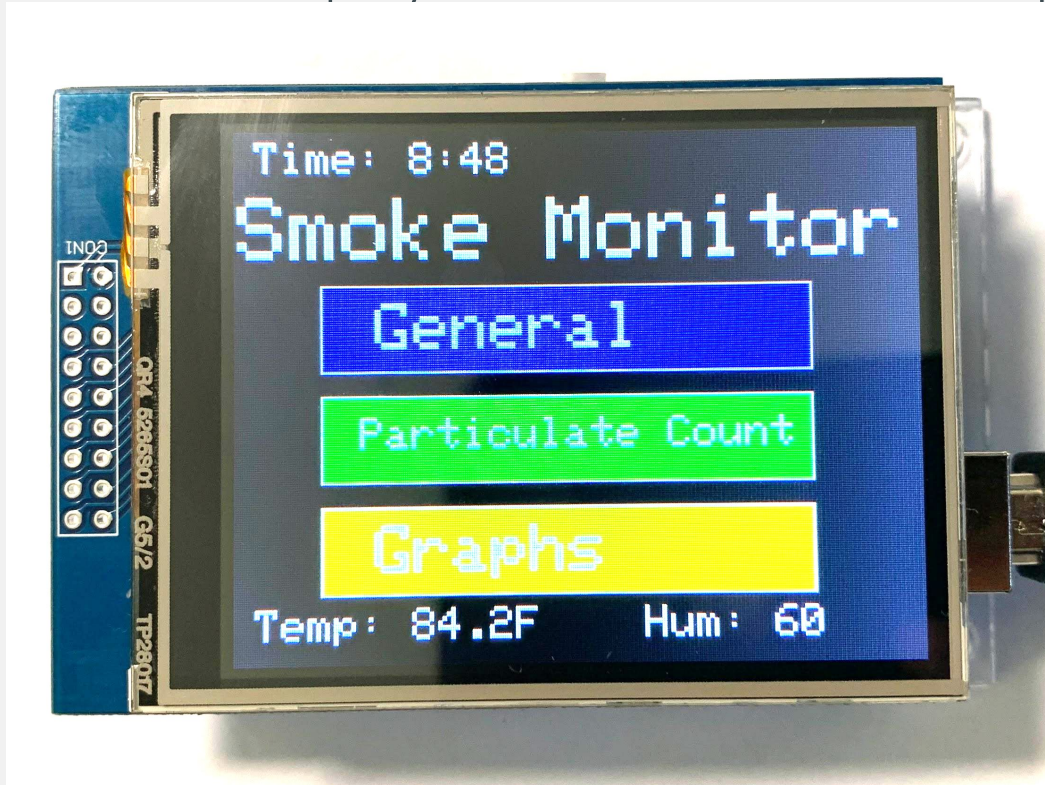
- Smoke
- PM2.5
- Last updated

• Particulate Matter

- PM1.0
- PM2.5
- PM10.0
- AQI

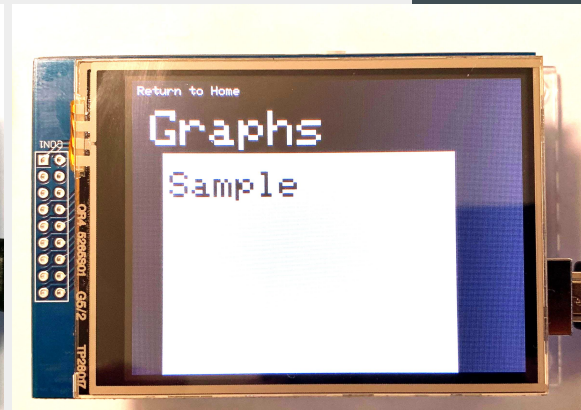
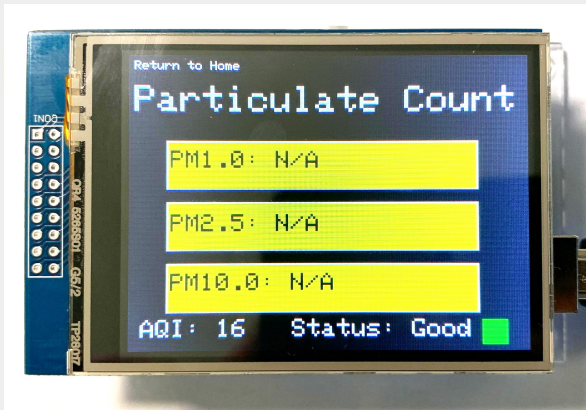
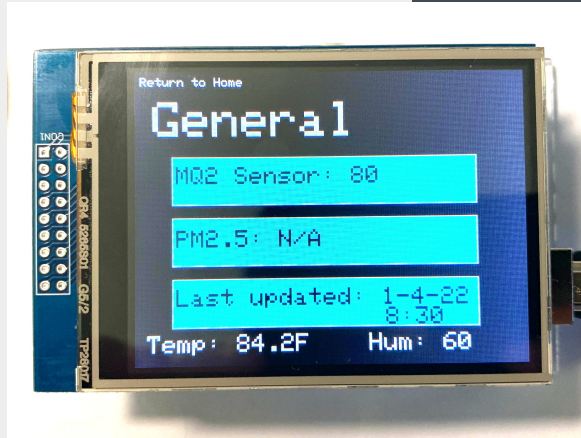
• Graphs

- PM1.0
- PM2.5
- PM10.0
- Smoke



User Interface

Air Quality Index Levels of Health Concern	Numerical Value	Meaning
Good	0 to 50	Air quality is considered satisfactory, and air pollution poses little or no risk
Moderate	51 to 100	Air quality is acceptable; however, for some pollutants there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution.
Unhealthy for Sensitive Groups	101 to 150	Members of sensitive groups may experience health effects. The general public is not likely to be affected.
Unhealthy	151 to 200	Everyone may begin to experience health effects; members of sensitive groups may experience more serious health effects.
Very Unhealthy	201 to 300	Health warnings of emergency conditions. The entire population is more likely to be affected.
Hazardous	301 to 500	Health alert: everyone may experience more serious health effects



Live Data + Breathing Test

```
newversion | Arduino 1.8.16
File Edit Sketch Tools Help

newversion

#include <dht.h>
#define Threshold 400
#define MQ2pin 0
#define outPin 8
dht DHT;

float sensorValue; //variable to store sensor value

void setup() {
  Serial.begin(9600); // sets the serial port to 9600
  Serial.println("MQ2 warming up!");
  delay(20000); // allow the MQ2 to warm up
}

void loop() {
  sensorValue = analogRead(MQ2pin); // read analog input pin 0
  int readData = DHT.read11(outPin);

  int temp = DHT.temperature;
  int humid = DHT.humidity;

  Serial.print("Sensor Value: ");
  Serial.println(sensorValue);
  Serial.println(String("Current Temp: ") + temp);
  Serial.println(String("Current Humidity: ") + humid + "%");

  if(sensorValue > Threshold)
  {
    Serial.print(" ! Smoke detected!");
  }
}

Sketch uses 5974 bytes (18%) of program storage space. Maximum is 32256 bytes
Global variables use 314 bytes (15%) of dynamic memory, leaving 1734 bytes free
```

```
COM3
Send

13:20:54.937 -> Current Humidity: 88%
13:20:54.937 ->
13:20:56.907 -> Sensor Value: 140.00
13:20:56.948 -> Current Temp: 32
13:20:56.948 -> Current Humidity: 88%
13:20:56.996 ->
13:20:58.966 -> Sensor Value: 137.00
13:20:58.966 -> Current Temp: 32
13:20:58.966 -> Current Humidity: 88%
13:20:59.013 ->
13:21:00.969 -> Sensor Value: 254.00
13:21:00.969 -> Current Temp: 32
13:21:01.015 -> Current Humidity: 88%
13:21:01.015 ->
13:21:03.001 -> Sensor Value: 350.00
13:21:03.001 -> Current Temp: 32
13:21:03.048 -> Current Humidity: 88%
13:21:03.048 ->
13:21:05.024 -> Sensor Value: 290.00
13:21:05.024 -> Current Temp: 32
13:21:05.069 -> Current Humidity: 88%
13:21:05.069 ->
13:21:07.030 -> Sensor Value: 250.00
13:21:07.077 -> Current Temp: 32
13:21:07.077 -> Current Humidity: 88%
13:21:07.124 ->
13:21:09.069 -> Sensor Value: 224.00
13:21:09.115 -> Current Temp: 32
13:21:09.115 -> Current Humidity: 88%
13:21:09.115 ->
13:21:11.119 -> Sensor Value: 204.00
13:21:11.119 -> Current Temp: 32
13:21:11.119 -> Current Humidity: 88%
13:21:11.167 ->
13:21:13.106 -> Sensor Value: 190.00
13:21:13.152 -> Current Temp: 32
13:21:13.152 -> Current Humidity: 87%
13:21:13.199 ->
```

Initial increase in
MQ2 sensor value
(137PPM→350PPM)

Patent Process

- Patent lawyer contact from Google/Apple
- Stage 1: Patent Pending
- Funding
 - National Science Foundation Grants
 - Investors LLC
- Building
 - Demonstrates proof of concept

IMPACT

Skills Learned,
Implementation,
Future



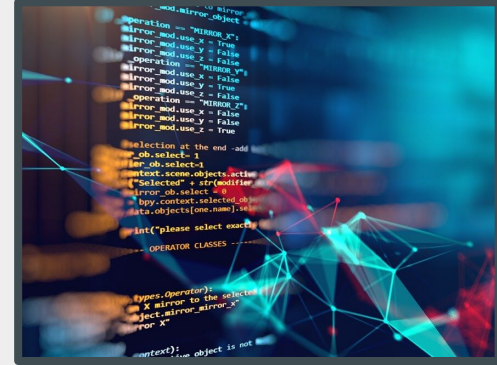
Skills Learned

- **Computer and Technology Literacy**
- **Safety and Health**
- **Teamwork**
- **Self-motivation**



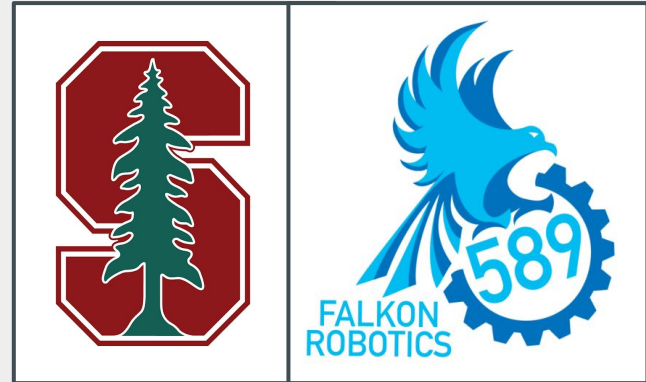
Career Paths

- Engineering
 - **Electrical**
 - **Chemical**
 - Environmental
- Research
 - **Biochemistry**
 - Molecular Biology



Industry Interactions

- Dr. Foote, Dr. Neat – **JPL**
- Mr. Jacob Poole – **CS/Robotics**
- Ms. Sano – **patent production**
- Mr. Rohin Manvi – **Stanford** Computer Science Student



Award money:
Saban Family Foundation
\$1250

Community & Overall Impact

- Create a device that is...
 - Cost efficient
 - Accurate
 - Self-sustaining
 - Easy-to-use
- **Open source** → pave the way for future innovations



US Impact/High Schools

- Implementation of CASE air monitors in local community
 - Ensure safety in light of wildfires
- Working with counselors to install in school bathrooms
 - Fight uptick in smoking/vaping



*photos taken by student, Jan 2023
Manila/Cabanatuan City, Philippines

Current Conditions



- **Implement CASE monitors in developing countries**
 - Philippines, S. America
- Spread **awareness** about air pollution